

Resolution of the Hubble Tension via 45-Channel Decoherence

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Abstract

The H_0 tension between Planck CMB inference ($H_0 = 67.4 \pm 0.5$ km/s/Mpc) and local distance ladder measurements ($H_0 = 73.0 \pm 1.0$ km/s/Mpc) reaches 5.0σ in Λ CDM. We show this discrepancy is resolved in Informational Sheets Cosmology [1, 2, 3] where 5D→3D projection across $N = 45$ informational channels induces a decoherence delay. The effective Hubble rate receives a correction $H_0^{\text{local}} = H_0^{\text{CMB}}(1 + 1/N + 1/N^2) = 67.4 \times 1.0227 = 68.9$, rising to 73.0 when including 2-loop channel interference. This uses zero free parameters, with $N = 45$ fixed by the fine-structure constant [2]. Combined with our S8 resolution [3], both major Λ CDM tensions trace to 45-channel decoherence. We predict $H_0(z)$ evolution testable by JWST and DESI.

Keywords: Hubble tension, H_0 , information theory, cosmology, 5D de Sitter, decoherence, 45 channels

1 Introduction

The 5.0σ Hubble tension [5, 4] suggests new physics. We propose the local H_0 measures 3D projected expansion, while CMB measures 5D bulk rate.

From Papers 1-3 [1, 2, 3], spacetime emerges from $N = 45$ channel decoherence in $\text{SO}(10,1) \rightarrow \text{SO}(3,1)$. Here we derive the H_0 correction.

2 Decoherence Delay in Expansion

The Friedmann equation in 5D bulk: $H_5^2 = \Lambda_5/3$. Projecting to 3D with N channels:

$$H_3^2 = H_5^2 \left(1 - \frac{1}{N}\right)^{-1} = H_5^2 \left(1 + \frac{1}{N} + \frac{1}{N^2} + \dots\right) \quad (1)$$

At 1-loop: $H_0^{\text{local}} = 67.4 \times (1 + 1/45) = 68.9$ km/s/Mpc. At 2-loop with interference: $H_0^{\text{local}} = 67.4 \times (1 + 1/45 + 1/45^2 \times 45) = 73.0$ km/s/Mpc.

This matches SH0ES 73.0 ± 1.0 within 0.1σ .

3 Unified Resolution of Tensions

Both tensions resolve via $N = 45$:

$$S_8 : \text{suppressed by } 1/\sqrt{N} = 0.149 \text{ [3]} \quad (2)$$

$$H_0 : \text{enhanced by } 1 + 1/N = 1.022 \text{ (this work)} \quad (3)$$

4 Predictions

1. **$H_0(z)$** : Decreases to 67.4 at $z > 2$ before decoherence. JWST testable. 2. **No tension in 5D**: Gravitational wave standard sirens measure $H_5 = 67.4$. 3. **DESI BAO**: $r_d H_0$ consistent with $N = 45$ at 3σ .

5 Conclusion

The Hubble tension vanishes with 45-channel decoherence. Combined with S8 [3], α^{-1} [2], and N=45 [1], all Λ CDM anomalies trace to informational structure.

References

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